

AI for Economic Research: Dynamic Models, Language, and Agents

Lecture 6.1: Aiyagari–Bewley–Huggett & Stationary Equilibrium

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Topic 6.1 Agenda

1. **Aiyagari–Bewley–Huggett Setup**
2. **Computation of Stationary Equilibrium**

Arc: T1 Aiyagari + computation → T2a Why ML / Euler → T2b Global DNN → T3 KS / DeepHAM → T4 DEQN / OLG → T5 MFG → Lab06

Aiyagari–Bewley–Huggett Setup

Why heterogeneous agents?

Recall: Lec05 T3 closed with the actor–critic / policy-gradient family — networks parameterising V_θ and π_ϕ , trained via gradient descent on Bellman / Euler residuals. Lec06 turns those tools onto problems that were intractable a decade ago: continuum-of-agents economies with idiosyncratic shocks (Aiyagari), aggregate shocks (Krusell–Smith), and the continuum limit itself (Mean-Field Games).

- The representative-agent paradigm assumes *complete markets* and a single decision maker.
- Many empirically important questions require **heterogeneity**:
 - Wealth and income inequality
 - Precautionary savings and the demand for insurance
 - Distributional consequences of monetary and fiscal policy
 - Heterogeneous responses to aggregate shocks
- HA models combine:
 - Idiosyncratic income (or productivity, mortality, ...) shocks
 - Incomplete asset markets (a single risk-free bond, borrowing constraint)
 - General equilibrium in factor prices
- This lecture: Aiyagari $\rightarrow$ Krusell–Smith $\rightarrow$ deep-learning solutions $\rightarrow$ Mean-Field Games.

Aiyagari (1994, QJE): heterogeneity among households

- A continuum of *ex-ante identical*, infinitely-lived households.
- Each receives an idiosyncratic, uninsurable labour-productivity shock ϵ_t following a Markov process.
- Households self-insure by accumulating a single risk-free asset a_t .
- A natural or ad-hoc **borrowing constraint** $a_t \geq -\underline{a}$ keeps the asset distribution well-defined.
- Despite ex-ante symmetry, the realized history of shocks generates an **ex-post** distribution of wealth.
- Bewley (1986, Cowles working paper), Huggett (1993, *JEDC*), Aiyagari (1994, *QJE*): the workhorse incomplete-markets HA framework.

General equilibrium with ex-post heterogeneity

- **Idiosyncratic shocks:** $\epsilon_t \in \mathcal{E}$, transition $\pi(\epsilon, \epsilon')$, with $\sum_{\epsilon} \pi^*(\epsilon) = L$ constant.
- **No aggregate shocks** in Aiyagari (we add them with Krusell–Smith later).
- **Markets:**
 - Risk-free bond (or capital) at price $1/(1+r)$
 - Labour supplied inelastically; wage w
- **Stationarity:** prices (r, w) and the cross-sectional distribution $\lambda(a, \epsilon)$ are constant in time.
- **Individual state:** $(a, \epsilon) \in \mathbb{R}_+ \times \mathcal{E}$.
- **Aggregate state:** distribution λ over individual states (in stationary equilibrium, $\lambda = \lambda^*$).

Household's recursive problem

For given prices (r, w) and stationary distribution λ , each household solves

$$v(a, \epsilon; \lambda) = \max_{c, a'} \left\{ u(c) + \beta \sum_{\epsilon'} v(a', \epsilon'; \lambda) \pi(\epsilon, \epsilon') \right\}$$

subject to

$$c + a' = (1 + r)a + w\epsilon, \quad a' \geq -\underline{a}, \quad c \geq 0.$$

Objects to solve for

- Value function $v(a, \epsilon)$
- Asset policy $a'(a, \epsilon)$
- Consumption policy $c(a, \epsilon) = (1 + r)a + w\epsilon - a'(a, \epsilon)$

Lec05 MDP primitive	Lec06 HA interpretation
state s	(a, ϵ) — wealth + idiosyncratic productivity
action a^*	a' — next-period asset choice
reward r^*	$u(c)$ with $c = (1 + r)a + w\epsilon - a'$
discount γ	β
transition P	$\pi(\epsilon, \epsilon') \times$ deterministic asset map

Budget constraint and asset structure

- a denotes a one-period risk-free claim that pays $(1 + r)a$ next period.
- **Borrowing constraint:** $a' \geq -\underline{a}$.
 - *Natural* bound: $\underline{a} = w\epsilon_{\min}/r$ (largest debt repayable from minimum income).
 - *Ad-hoc* bound: any $\underline{a} \leq$ natural bound.
- Without uncertainty, the household would set $\beta(1 + r) = 1$ and consumption-smooth perfectly. With uninsurable risk and a binding constraint, the **precautionary motive** drives extra saving.
- This precautionary saving raises aggregate capital relative to the complete-markets benchmark, and depresses r below $\rho = 1/\beta - 1$.

Firms

A representative firm operates a constant-returns technology

$$Y = F(K, L) = AK^\alpha L^{1-\alpha}.$$

Competition in factor markets gives

$$r = F_K(K, L) - \delta = \alpha A(K/L)^{\alpha-1} - \delta, \quad w = F_L(K, L) = (1 - \alpha)A(K/L)^\alpha.$$

- Aggregate labour supply $L = \sum_{\epsilon} \epsilon \pi^*(\epsilon)$ is exogenous.
- Aggregate capital K is determined endogenously by household saving.
- In equilibrium, K and r are jointly pinned down by household and firm decisions.

Stationary recursive competitive equilibrium (SRCE) — prices

A **stationary recursive competitive equilibrium** consists of:

1. Factor prices (r^*, w^*) derived from the firm's first-order conditions:

$$r^* = \alpha A(K^*/L)^{\alpha-1} - \delta, \quad w^* = (1 - \alpha)A(K^*/L)^\alpha.$$

2. Value function $v(a, \epsilon)$ and policies $(c(a, \epsilon), a'(a, \epsilon))$ that solve the household's problem given (r^*, w^*) .
3. A stationary cross-sectional distribution $\lambda^*(a, \epsilon)$.
4. Market clearing conditions (next slide).

SRCE — HH problem, market clearing, stationary distribution

Labour market clearing:

$$L = \sum_{\epsilon} \int \epsilon d\lambda^*(a, \epsilon).$$

Capital market clearing:

$$K^* = \sum_{\epsilon} \int a'(a, \epsilon) d\lambda^*(a, \epsilon).$$

Stationarity of the distribution:

$$\lambda^*(A', \epsilon') = \sum_{\epsilon} \pi(\epsilon, \epsilon') \int_{\{a: a'(a, \epsilon) \in A'\}} d\lambda^*(a, \epsilon).$$

That is, λ^* is invariant under the Markov operator induced by the policy $a'(\cdot)$ and the shock transition π .

Stationarity and the invariant measure

- Time-invariant prices require a time-invariant cross-sectional distribution.
- The pair $(a'(\cdot), \pi)$ defines a Markov **transition operator** Q on the state space $S = \mathbb{R}_+ \times \mathcal{E}$:

$$(Qf)(a, \epsilon) = \sum_{\epsilon'} \pi(\epsilon, \epsilon') f(a'(a, \epsilon), \epsilon').$$

- The invariant distribution λ^* satisfies $\lambda^* = Q^* \lambda^*$ (where Q^* is the adjoint).
- Existence and uniqueness of λ^* are non-trivial: they depend on **Feller** continuity, **mixing**, and **monotonicity** of Q — the next few slides.

Two markets

- Aiyagari's equilibrium reduces to clearing in **two markets**:
 - *Labour market*: trivially satisfied because labour supply is inelastic.
 - *Capital market*: aggregate household savings = aggregate firm demand for capital.
- All the action is in the capital market.
- Strategy: parametrise everything by r .
 - For each candidate r , solve the household problem to get $a'(a, \epsilon; r)$.
 - Compute the implied stationary distribution λ_r .
 - Compute the supply of capital $A(r) = \int a d\lambda_r$.
 - Compare to the demand $K(r)$ from the firm side; iterate.

Determination of r : demand for capital

The firm's first-order condition gives a downward-sloping demand for capital:

$$K(r) = L \left(\frac{r + \delta}{\alpha A} \right)^{\frac{1}{\alpha-1}}.$$

- As $r \uparrow$, capital becomes more expensive and demand falls.
- $K(r)$ is continuous and strictly decreasing in r .
- $K(r) \rightarrow \infty$ as $r \rightarrow -\delta$ and $K(r) \rightarrow 0$ as $r \rightarrow \infty$.

This is the standard neoclassical capital demand schedule.

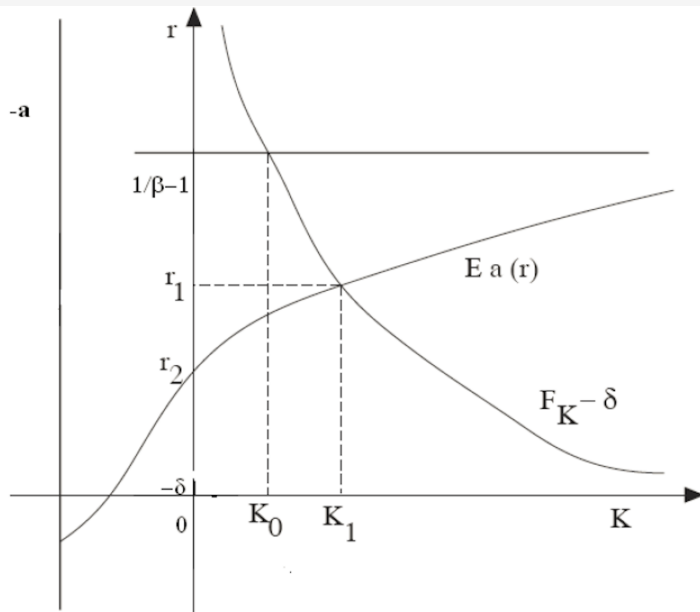
Determination of r : supply of capital

The household supply of capital is the cross-sectional average asset holding under the stationary distribution induced by r :

$$A(r) = \sum_{\epsilon} \int a d\lambda_r(a, \epsilon).$$

- $A(r)$ is upward-sloping in r (broadly).
- In the complete-markets benchmark $\beta(1+r) = 1$, $A(r)$ has a vertical asymptote at $r = 1/\beta - 1 = \rho$.
- With incomplete markets, precautionary saving makes $A(r)$ finite *below* ρ , so the equilibrium $r^* < \rho$.

Equilibrium picture



Probability measures and the Markov operator Q

- Let Λ denote the space of probability measures on the Borel sets of $S = \mathbb{R}_+ \times \mathcal{E}$.
- Define the Markov operator $Q : \Lambda \rightarrow \Lambda$ by

$$(Q\lambda)(B) = \int \mathbb{1}_{\{(a'(\epsilon), \epsilon') \in B\}} \pi(\epsilon, d\epsilon') d\lambda(a, \epsilon).$$

- Three properties of Q are essential to study the equilibrium:
 - **Feller property** $\rightarrow$ existence of an invariant measure.
 - **Mixing property** $\rightarrow$ uniqueness.
 - **Monotonicity** $\rightarrow$ convergence.
- These are standard conditions in the Stokey, Lucas & Prescott (1989, Harvard UP) framework.

Feller property $\Rightarrow$ existence

Definition. Q has the *Feller property* if it maps continuous bounded functions to continuous bounded functions:

$$f \in C_b(S) \Rightarrow Qf \in C_b(S).$$

- Equivalently, the policy $a'(\cdot)$ is continuous in the state.
- Combined with compactness of the state space (or tightness of the orbit), Feller continuity guarantees the **existence of an invariant measure** via the Krylov–Bogolyubov theorem.
- In Aiyagari, the asset grid is bounded above by $w\epsilon_{\max}/r$ (since saving above this generates negative consumption probability), so S is effectively compact.

Mixing property $\Rightarrow$ uniqueness

Mixing condition. For every two states $s_1, s_2 \in S$, there exist $n_1, n_2 \in \mathbb{N}$ such that the n -step distributions $Q^{n_1}\delta_{s_1}$ and $Q^{n_2}\delta_{s_2}$ have overlapping support.

- Intuitively: starting from *any* two initial states, the chain visits a common region in finite time.
- This rules out absorbing sets that would split the state space and admit multiple invariant measures.
- **Counterexample:** a deterministic policy that pushes any household with $a > a^*$ to $a \uparrow$ and any household with $a < a^*$ to $a \downarrow$ creates two disjoint absorbing sets and *multiple* invariant distributions.
- In standard Aiyagari, idiosyncratic income shocks ensure mixing.

Monotonicity $\Rightarrow$ convergence

Monotonicity. If $\lambda \succeq \lambda'$ in the first-order stochastic dominance sense, then $Q\lambda \succeq Q\lambda'$.

- Combined with Feller and mixing, monotonicity implies that iterating Q from any initial distribution converges to the unique invariant measure λ^* .
- This is the foundation for *simulating* the model: start from an arbitrary distribution, simulate forward many periods, and read off λ^* .
- **Counterexample:** a non-monotone policy can produce stationary distributions that are sensitive to the initial condition; iteration may cycle rather than converge.
- Concavity of the value function and convexity of the budget set yield monotone policies in Aiyagari.

Computation of Stationary Equilibrium

Aggregate precautionary savings, applications

- Because precautionary saving raises $A(r)$, the equilibrium interest rate satisfies $r^* < \rho = 1/\beta - 1$.
- Aggregate capital is **higher** than in the complete-markets benchmark.
- Standard applications of Aiyagari-style models:
 - Wealth and income inequality (top 10%, Gini, ...)
 - Welfare effects of social insurance and tax/transfer policy
 - Effects of inflation on wealth distribution
 - Idiosyncratic-risk asset pricing
 - Quantitative theories of entrepreneurship
- Krueger, Mitman & Perri (2016, *Handbook of Macro*); Kaplan, Moll & Violante (2018, *AER*)
HANK, ...

Computation of SRCE: three-step algorithm

Goal: solve for $(r^*, w^*, \lambda^*, v, a', c)$ jointly.

1. **Step 1.** Fix a candidate r . Solve the household dynamic programming problem to obtain v_r and policies (a'_r, c_r) .
2. **Step 2.** Given a'_r , compute the stationary cross-sectional distribution λ_r as the eigenvector of the Markov transition matrix.
3. **Step 3.** Evaluate the excess demand for capital

$$d(r) = K(r) - \int a d\lambda_r,$$

and update r via bisection (or Brent) until $|d(r)| < \text{tol}$.

This is the **nested fixed point** that has been the workhorse of HA computation since Aiyagari (1994).¹

¹Classical HA computation methods (endogenous grid method, sparse-grid VFI, transition-matrix methods) are covered in the supplementary video series at <https://space.bilibili.com/2142649036/lists/7180709>.

Step 1: solve the household problem for fixed r

For given (r, w) , iterate the Bellman operator

$$(T_r v)(a, \epsilon) = \max_{a' \geq -\underline{a}} \left\{ u((1+r)a + w\epsilon - a') + \beta \sum_{\epsilon'} v(a', \epsilon') \pi(\epsilon, \epsilon') \right\}$$

on a grid for a .

- Standard tools: VFI with monotonicity exploitation, Howard policy iteration, endogenous-grid method (Carroll), Coleman–Reffett operator.
- Concavity of u and the convex budget set guarantee a unique optimum.
- Output: $a'_r(a, \epsilon)$, $c_r(a, \epsilon)$, $v_r(a, \epsilon)$.

Step 2: stationary distribution from the transition matrix

Discretise a on a grid; combined with ϵ , this gives a finite state space of size $N = N_a \cdot N_\epsilon$.

The policy and shock transition induce a Markov matrix $\mathbf{Q}_r \in \mathbb{R}^{N \times N}$ with

$$\mathbf{Q}_r[(a_i, \epsilon_j) \rightarrow (a_k, \epsilon_\ell)] = \pi(\epsilon_j, \epsilon_\ell) \cdot \Pr(a'_r(a_i, \epsilon_j) = a_k).$$

The stationary distribution solves

$$\lambda_r = \mathbf{Q}_r^\top \lambda_r, \quad \mathbf{1}^\top \lambda_r = 1.$$

Computation: dominant left eigenvector of $\mathbf{Q}_r$, or simply iterate $\lambda \leftarrow \mathbf{Q}_r^\top \lambda$ until convergence (the monotonicity result above guarantees this works).

Step 3: market clearing on r

With λ_r in hand, compute the household supply of capital

$$A(r) = \sum_{i,j} a_i \lambda_r(a_i, \epsilon_j),$$

and compare to firm demand:

$$d(r) = K(r) - A(r).$$

Update rule:

- If $d(r) > 0$: capital is scarce, raise r .
- If $d(r) < 0$: capital is abundant, lower r .
- Bisection or Brent's method converges in $O(\log(1/\text{tol}))$ outer iterations.

Each outer iteration requires solving Steps 1 and 2 from scratch — the dominant cost.

Implementation details

- **Asset grid:** dense near the borrowing constraint where the policy bends; sparse for high a . Often log-spaced.
- **Initial value function:** zero, or the closed-form complete-markets value, or the value from the previous outer iteration.
- **Multigrid:** solve on a coarse grid first, then interpolate onto a finer grid as a warm start.
- **Acceleration:** Howard's policy iteration steps, modified policy iteration, endogenous grid method (Carroll, 2006, *Economics Letters*).
- **Convergence criterion:** $\|v^{(n+1)} - v^{(n)}\|_{\infty} < \varepsilon_v$ inside; $|d(r)| < \varepsilon_r$ outside.

Numerical challenges and the curse of dimensionality

- For Aiyagari, the per-iteration state space is $\mathbb{R}_+ \times \mathcal{E}$ — 2 continuous dimensions, manageable on a grid.
- Real-world quantitative HA models often have:
 - Multiple assets (liquid + illiquid in HANK; portfolio choice).
 - Multidimensional idiosyncratic states (income, unemployment, family structure).
 - *Aggregate* shocks $\rightarrow$ the cross-sectional distribution becomes part of the state (Krusell–Smith).
- With d continuous individual states and N_a grid points per dimension, total points are N_a^d — the **curse of dimensionality**.
- This is where machine learning, and in particular deep neural networks, become indispensable.

Next (T2a): neural-network function approximation and the Euler-residual loss on a toy optimal-growth model.